

KRISHKUMAR MURUGAN

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EDUCATION

Clark University	May 2025
Master of Science, Computer Science	GPA-3.9
Vel Tech Institute of Science and Technology , Chennai, India	June 2023
Bachelor of Technology, Computer Science and Engineering	GPA-3.7

SKILLS

Technical skills	C, C++, Python, Java, HTML, SQL, Microsoft Office Suite(Word, PowerPoint, Excel)
Packages	NumPy, Pandas, Matplotlib, Pytorch, Tensorflow
Visualization Tools	IBM Business Intelligence tool, Tableau

PROJECTS

- OnMart Superstore – E-commerce Application using MERN Stack** August 2024-December 2024
- Improved user experience and performance using React.js for the frontend, Express.js for the backend, and MongoDB as the database, delivering seamless features.
 - Increased platform scalability by 40%** and reduced deployment time by 20% through the use of AWS EC2 and Docker containers for deployment, with GitHub ensuring efficient version control and maintenance.
 - Achieved 95% test coverage** of core functionalities by conducting rigorous functional and non-functional testing and automating testing processes using Selenium, ensuring availability, performance, and user satisfaction.

- Report Summarization using Flan T5 Model** May 2024-July 2024
- Built a report summarizer using the Flan T5 model with **Pytorch and Tensorflow** libraries.
 - Fine-tuned the model with the Samsum dataset (16k messenger-like conversations with summaries), creating a more robust algorithm that processed data points seamlessly.
 - Achieved a **ROUGE score of 46.77**, indicating a high level of summarization accuracy.

- Exploratory Data Analysis Portal** January 2024-May 2024
- Enhanced user experience by 30% by designing a dynamic and responsive frontend using React, enabling seamless interactions and real-time data visualizations for an interactive EDA portal.
 - Reduced data processing time by 15%** and improved database query performance by 40% by implementing a Flask-based API with optimized routing and integrating MongoDB for efficient large-scale dataset storage and retrieval.
 - Strengthened portal security by 25%** and ensured zero downtime during deployment by implementing JWT-based authentication and collaborating with a team of three to deliver a robust and scalable end-to-end solution.

EXPERIENCE

- IoT Consensus Mechanism - Research Internship** August 2022-February 2023
- Prof. Zachary Auhl, Graduate Researcher, Computer Science & Information Technology
La Trobe University, Australia.*

- Developed Multivariate Anomaly Detection with Generative Adversarial Networks (MAD-GAN), lightweight **machine learning algorithm** to run on Raspberry Pis for sensor data anomaly detection.

- AICTE Artificial Intelligence-Machine Learning Intern** October 2021-December 2021
- Dr. Syed Fiaz, EduSkills Instructor
Amazon Web Services Academy.*

- Created forecast with time series data and used Amazon Lex to create a chatbot using Natural Language Processing and achieved an 84% efficiency.
- Implemented a Machine Learning pipeline** with Amazon SageMaker and created a forecast using time series data with Amazon Forecast.

RESEARCH PAPERS

- [Prediction and Forecasting of Air Pollution Level Using ARIMA and SARIMA based on Time-series Data](#)
- [Intelligent Mart System Using NLP For Smart Cities March 2022](#)